<u>Definition</u>: State i is said to be <u>recurrent</u> if, starting from i, the chain will return to i with probability 1.

<u>Definition</u>: If state i is not recurrent it is <u>transient</u>. (There is a positive probability of escaping i, never to return again!)

$$
\begin{array}{c|ccccc}
 & 1 & 2 & 3 & 4 & 5 \\
\hline
1 & 1/3 & 0 & 2/3 & 0 & 0 \\
2 & 1/4 & 1/8 & 1/8 & 1/4 & 1/4 \\
3 & 1/2 & 0 & 1/2 & 0 & 0 \\
4 & 0 & 2/3 & 0 & 1/3 & 0 \\
5 & 1/4 & 3/4 & 0 & 0 & 0
\end{array}
$$

$\{1,3\}, \{2,4,5\}$

State 2 is transient since it is possible to move from 2 to 1, after which there is no way back!

Note that:

① It is possible to move between classes but only in one direction!

Note that: $\{1,3\}, \{2,4,5\}$

② Without looking back at the matrix, we know that 4 and 5 must be transient since they communicate with 2 and, from 2, we can leave the $\{2,4,5\}$ class entirely!

$\boxed{\text{"Transience is a class property."}}$

Note that: $\{1,3\}, \{2,4,5\}$

③ States 1 and 3 are recurrent.

$$
\begin{array}{c|ccccc}
 & 1 & 2 & 3 & 4 & 5 \\
\hline
1 & 1/3 & 0 & 2/3 & 0 & 0 \\
2 & 1/4 & 1/8 & 1/8 & 1/4 & 1/4 \\
3 & 1/2 & 0 & 1/2 & 0 & 0 \\
4 & 0 & 2/3 & 0 & 1/3 & 0 \\
5 & 1/4 & 3/4 & 0 & 0 & 0
\end{array}
$$

<u>Question</u>: Isn't there a chance of going from 1 to 3 and then repeatedly going from 3 to 3 forever?

i.e. How do we know that 1 (for example) is recurrent?

Example:

$$\mathbb{P} = \begin{array}{c c} & \begin{array}{cc} 1 & 3 \end{array} \\ \begin{array}{c} 1 \\ 3 \end{array} & \begin{bmatrix} 0.02 & 0.98 \\ 0.01 & 0.99 \end{bmatrix} \end{array}$$

Starting at 1, will show that we will return with probability 1.

Will show for general $p_{11}, p_{13}, p_{31}, p_{33} > 0$

(Thus all are < 1.)

$P(\text{return to } 1 \mid X_0 = 1)$

$= P(\underline{\underline{\text{first}}} \text{ return in } 1 \text{ step}) \leftarrow p_{11}$

$+ P(\underline{\underline{\text{first}}} \text{ return in } 2 \text{ steps}) \leftarrow p_{13} \cdot p_{31}$

$+ P(\underline{\underline{\text{first}}} \text{ return in } 3 \text{ steps}) \leftarrow p_{13} \cdot p_{33} \cdot p_{31}$

$+ P(\underline{\underline{\text{first}}} \text{ return in } 4 \text{ steps}) \leftarrow p_{13} \cdot p_{33}^{2} \cdot p_{31}$

$\vdots$

$$P(\text{return to } 1 \mid X_0 = 1)$$

$$= p_{11} + \sum_{n=0}^{\infty} p_{13} \cdot p_{33}^{n} \cdot p_{31}$$

$$= p_{11} + p_{13}\, p_{31} \sum_{n=0}^{\infty} p_{33}^{n} \quad \text{(Note that } 0 < p_{33} < 1.\text{)}$$

$$= p_{11} + p_{13}\, \cancel{p_{31}} \cancel{\frac{1}{1 - p_{33}}} = p_{31}$$

$$= p_{11} + p_{13} = 1 \checkmark$$

⊛ Transience and recurrence are both "class properties".

We, therefore refer to entire communication classes as either transient or recurrent.

<u>Definition</u>: If all states of a Markov chain communicate, the chain is said to be <u>irreducible</u>.

- For an irreducible Markov chain, the entire chain must be transient or recurrent.

- A Markov chain on a finite state space must have at least one recurrent state.

- Thus, an irreducible Markov chain on a finite state space has all states recurrent and is therefore a recurrent chain.